Supplementary Information

An open-source, 3D-printed auger powder doser designed with generative artificial intelligence

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S1 Bill of materials

Table S1 lists the parts for one channel, plus the items a platform needs only once whatever its number of channels. Prices are single-unit US dollars as of September 2026 and should be checked at order time; vendor links are kept in `hardware/` in the repository (`https://github.com/vertical-cloud-lab/powder-doser`).

Table S1: Bill of materials. Printed parts are excluded (about 260 g of PLA per channel, under $10), as are the custom circuit board, which can be replaced by a breadboard, and the analytical balance (A&D HR-100A, 0.1 mg readability), which most laboratories already own.

Part	Qty	Price (USD)	Source
Per channel			
NEMA-11 bipolar stepper, 0.67 A, 10 N·cm (11HS18-0674S)	1	14	StepperOnline
Pololu Tic T500 stepper controller	1	33	Pololu #3135
Pololu 33 V shunt regulator (clamps stepper back-voltage)	1	15	Pololu #3776
12 V push–pull solenoid, JF-0530B type (tapping)	1	7.50	Adafruit #412
DRV8871 H-bridge breakout (solenoid driver)	1	7.50	Adafruit #3190
ERM coin vibration motor, 10 mm	1	1.95	Adafruit #1201
DRV2605L haptic driver breakout (I^2C)	1	7.95	Adafruit #2305
MG996R-class metal-gear servo (tilt)	2	24	generic
Fasteners (M3), bulk capacitors	—	5	generic
Shared (one per platform)			
Raspberry Pi Pico W	1	6	Raspberry Pi
Waveshare Pico-2CH-RS232 board (balance interface)	1	9	Waveshare 19979
Mean Well GST60A12-P1J 12 V/5 A supply, cord, and pigtail	1	26	Digi-Key
Pololu D24V22F5 5 V buck regulator	1	19	Pololu #2858
Headers and wiring	—	10	generic
Total, single-channel platform		**≈185**	

S2 Construction guide

Table S2 lists the printed parts of the current doser and where to download each one. Fig. S1 shows every part of the mechanical assembly, numbered, with its bill of materials, and Fig. S2 shows the build in 19 steps. Both come from a CAD assembly of the team's part files, which is also a public Onshape document (`https://cad.onshape.com/documents/ae9f107d3972fc9d390e541f`); the same steps are animated in the repository README. The wiring guide and firmware are in `hardware/` in the repository. The build, in outline (step numbers refer to Fig. S2, item numbers to Fig. S1):

1. **Print** the parts in Table S2 in PLA (0.2 mm layers, four perimeters, 25% infill). Most parts need support material. Print the auger upright.

2. **Base** (steps 1–5): screw the baseplate to a flat board or bench top (the CAD assumes 38 mm, the depth its front legs are made for), fit the two servos into its posts, and fit a 14-tooth pinion on each servo.

3. **Tilt stage** (steps 6–7): mount the mounting plate on the hinge with M5 screws and locknuts, so that the servo pinions mesh with the plate's 28-tooth gears (2:1).

4. **Drive** (steps 8–10): bolt the NEMA-11 stepper to its plate, fit the 20-tooth pinion, and bolt the tap-collar base to the plate.

5. **Auger** (steps 11–17): slide one bracket and then the tap collar onto the tube from the outlet end, lower the auger onto the plate so that its 44-tooth gear meshes with the stepper pinion (2.2:1), slide the second bracket on from the cap end, and tighten the bracket screws and the tap-collar clamp. The collar's ears rest against the raised stop on its base, so the collar stays put while the tube turns inside it.

6. **Solenoid** (step 18): screw the solenoid to the tap collar so that its plunger strikes the tube. (A vibration motor was also planned for the collar but was not used in this work.)

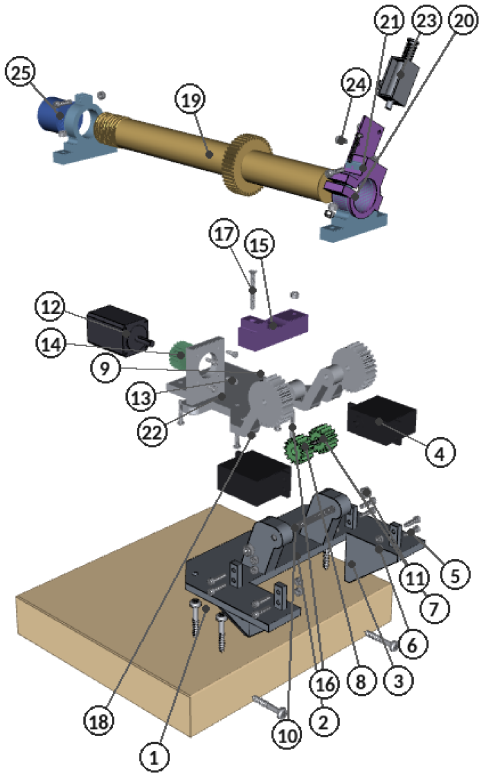
7. **Electronics:** wire the Pico W, Tic T500 (serial), DRV8871, DRV2605L (I^2C), the RS-232 board (to the HR-100A balance), and the servos as shown in the schematic in `hardware/test-module/`, then flash the MicroPython firmware.

8. **Fill** (step 19): with the auger standing upright in the printed filling stand, unscrew the cap, pour the powder into the upper end of the tube, and screw the cap back on. There are no loading slots; the tube is its own reservoir. For small or costly samples, one of the two smaller augers in Table S2 can take its place.

9. **Checkout:** run the firmware self-test (rotate, tap, tilt, tare, and read), then the test protocols listed in the main text.

Table S2: Printed parts of the current doser and their design files. Every part except the tap-collar base was modelled by the team in Fusion 360, most of them as recreations of AI-modelled parts, and is shared publicly with downloads enabled. The tap-collar base is still the AI-modelled part, written by the coding agent in May 2026 (pull request #51). The auger, its cap, and the stepper pinion were on the rig for every test in the main text. STEP exports of the Fusion 360 parts are also in the repository (`cad/full-assembly/components/fusion-step/`, pull request #170), and the AI-modelled originals are in `cad/`.

Part	Design file
Auger: threaded storage tube with 44-tooth gear	`https://a360.co/4y1oz2H`
Screw-on cap	`https://a360.co/4w9kRE5`
Stepper pinion, 20 teeth	`https://a360.co/4yqSHFz`
Auger brackets (two)	`https://a360.co/46XtYN1`
Tap collar	`https://a360.co/4AIIgyz`
Tap-collar base (AI-modelled)	`mount_plate.step`, pull request #51
Mounting plate	`https://a360.co/4xXUj8L`
Baseplate	`https://a360.co/4AOA7sI`
Servo pinions, 14 teeth (two)	`https://a360.co/4dcGSdF`
Accessories	
Auger filling stand	`https://a360.co/4yqMPfp`
Smaller auger, about 3.3 mL	`https://a360.co/4hxzCu5`
Smaller auger, about 9.2 mL	`https://a360.co/47qXATc`



Bill of materials

Exploded view

No.	Part	Qty	Type
1	Mounting board	1	yours
2	Baseplate	1	printed
3	#10×1¼ in pan head wood screw	6	fastener
4	Servo	2	purchased
5	M3×14 socket head screw	10	fastener
6	M3 nylon-insert locknut	16	fastener
7	Servo pinion	2	printed
8	M3×10 socket head screw	2	fastener
9	Mounting plate	1	printed
10	M5×45 button head screw	2	fastener
11	M5 nylon-insert locknut	2	fastener
12	Stepper motor	1	purchased
13	M2.5×8 socket head screw	4	fastener
14	Stepper pinion	1	printed
15	Tap-collar base	1	printed
16	M3×25 button head screw	1	fastener
17	M3×30 flat head screw	1	fastener
18	M3 hex nut	1	fastener
19	Auger	1	printed
20	Bracket	2	printed
21	Tap collar	1	printed
22	M3×20 button head screw	5	fastener
23	Solenoid	1	purchased
24	M3×5 socket head screw	2	fastener
25	Auger cap	1	printed

Figure S1: Exploded view of the current design with every item of its mechanical bill of materials numbered: printed parts, purchased parts, and fasteners (McMaster-Carr part numbers in `cad/full-assembly/BOM.md`). Item 1 is whatever the doser is screwed to; the CAD uses a board 38 mm thick, the depth the baseplate's front legs are made for. The printed parts are the files in Table S2; the stepper, servos, and solenoid are simplified models made from their datasheets. Electronics are listed in Table S1. The CAD assembly was built by an AI agent from the team's part files, working through Onshape's programming interface, and rendered from the same files (pull request #170).

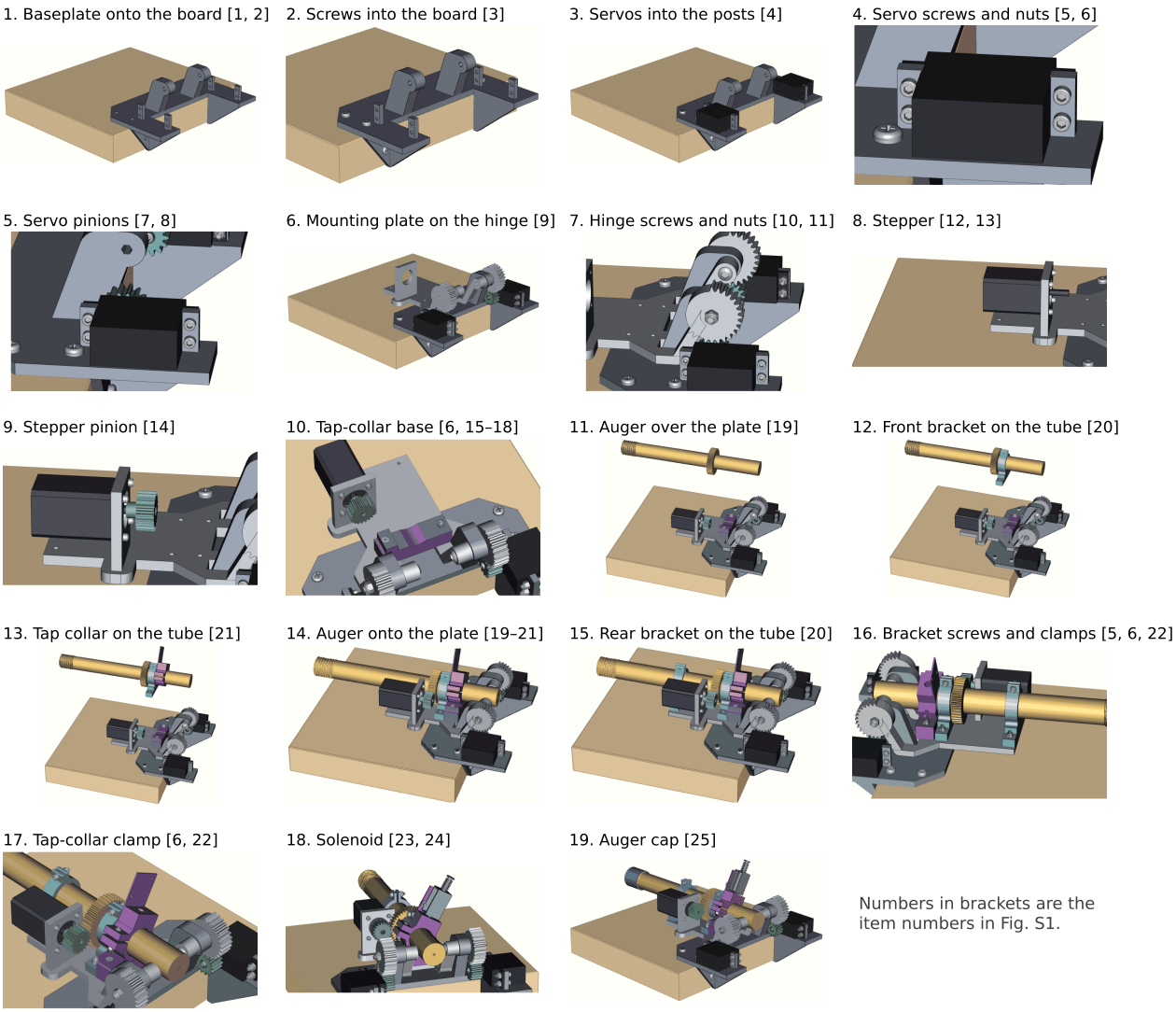


Figure S2: The build in 19 steps, as stills from the end of each step of the assembly animation in the repository. Bracketed numbers are the item numbers of Fig. S1. This order passes the CAD assembly's collision checks; it is one workable order rather than a record of how the test rig was built.

S3 Auger and outlet geometry

Fig. S3 shows the AI-modelled geared auger and its four printed outlet variants, made early in the project. The auger used in the tests is the team's later Fusion 360 part, shown as a cut-away in Fig. 1c of the main text. Like the Type 4 variant, it ends in a tapered outlet with a 3 mm exit hole. Table S3 compares it feature by feature with the first AI-modelled geared auger. In June 2026 the coding agent also modelled, at our request, a threaded storage version with a screw-on cap and the flight confined to the outlet third. The tested auger follows that design, so the team's own changes were the gear pair, a uniform flight at a slightly longer pitch, and the cap thread and size.

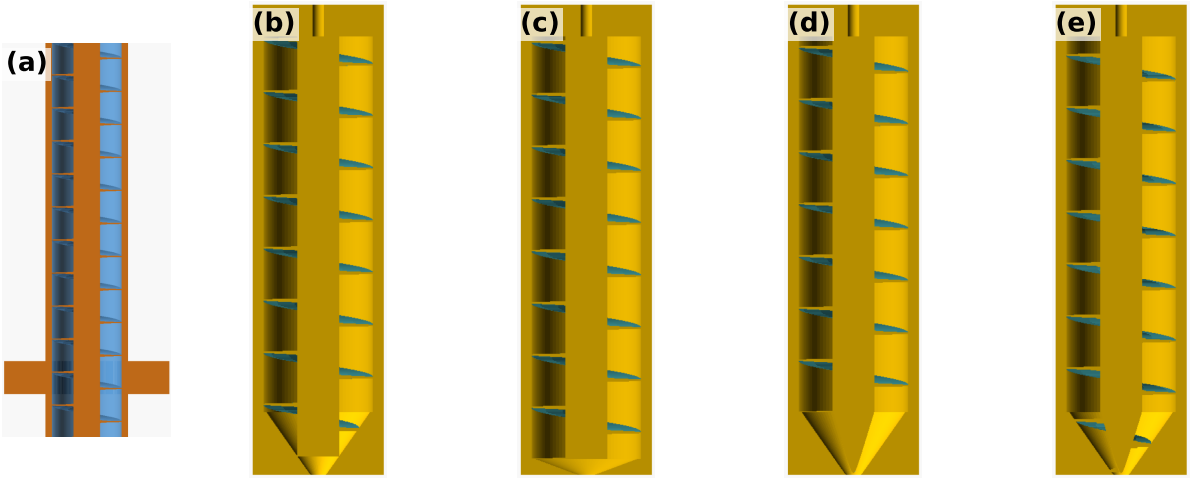


Figure S3: (a) Cross-section of the AI-modelled geared auger: the 8 mm central core, the single-start helical flight (10 mm pitch) running the full length of the 21 mm bore, and the external gear band. (b–e) Cross-sections of the four outlet variants (Types 1–4), which differ in exit constriction and chamfer. All are parametric OpenSCAD models (`cad/auger-geared/` in the repository) written by AI coding agents under human review; no conventional CAD software was used to design them. The auger used in the tests is the team's later Fusion 360 part (main text, Fig. 1c). Its flight is 0.5 mm thick with a 10.4 mm pitch and covers only the third of the tube nearest the outlet, the tube is filled through a screw-on cap rather than through loading slots, and its gear has 44 teeth rather than 48 (Table S3).

S4 AI-assisted CAD workflow

Fig. S4 shows how the AI-modelled parts were made. A team member opened a GitHub issue for one part, often with an annotated sketch (Fig. S4a). The GitHub Copilot coding agent wrote the part as CadQuery or OpenSCAD code and opened a pull request. Scripts in its session rendered views, exported STL and STEP files, and checked that the part was one closed solid (pull request #53) that did not overlap its neighbours (pull request #66). We reviewed the renders and replied in the pull request, often with marked-up screenshots. Each reply started a new agent session, with a median length of 11 min. We printed accepted parts, fitted them, and tested them on the doser, and problems went back to the agent. Issue #34 summed up the loop as "AI design → printing → testing → feedback → AI design". From June 2026, some parts, including the final tap collar, were modelled instead by chatting with the Zookeeper agent in Zoo Design Studio.

Each request set out how the part met its neighbours:

- mating parts and their files (the first auger for the bracket, pull request #47; the stepper motor of issue #24 for the pinion, issue #48);
- holes and fasteners (clamp screw and mounting holes, issue #46; M5 holes on a square, pull request #66);
- bores and clearances (the auger diameter "with some tolerance", issue #46);
- gear ratios and axes (2:1 servo gear, issue #65; hinge axis through the dosing point, issue #56);
- the face to print on (issue #46);

Table S3: The AI-modelled geared auger compared with the auger used in every test. AI-modelled values are the parameters of the OpenSCAD model that the coding agent wrote in May 2026 (`cad/auger-geared/auger-core.scad`, `archimedes-auger-geared.scad`, `gear-teeth.scad`, and `stepper-pinion.scad`; pull request #49). Tested values were measured on the team's Fusion 360 files (`threaded-auger-final.stl`, `cap-final.stl`, and `stepper-pinion.step`; pull request #170). Lengths along the tube are measured from the outlet. Flight thickness is given along the axis for both parts; the 2 mm in the AI code is the width of the blade across a section normal to the axis. Values and their sources are in `paper/figures/data/auger_comparison.csv`.

Feature	AI-modelled auger	Tested auger	What changed
Overall length	250 mm	250 mm	Unchanged
Outer diameter	25 mm	25 mm	Unchanged
Bore	21 mm	21 mm	Unchanged
Wall thickness	2 mm	2 mm; 1 mm under the cap thread	Thinner under the thread[a]
Core	8 mm across, 3.5–244 mm from the outlet	8 mm across, 0–83 mm; tapers to 0.8 mm at the exit	Outlet third only; carried down to the exit[a]
Flight thickness	2 mm across the blade in the code; 0.3–0.8 mm along the axis, thinnest at the wall	0.5 mm along the axis, the same at every radius	Uniform; about the same at mid-radius (0.44 mm in the AI model)
Flight pitch	10 mm	10.4 mm	4% longer
Flighted length	3.5–244 mm (24 turns)	0–84 mm (8 turns)	Outlet third only[a]
Number of starts	One, right-handed	One, right-handed	Unchanged
Outlet and exit hole	12 mm cone, 20 to 3 mm across; the core stops where it meets the cone, 3.5 mm above the exit	12 mm cone, 21 to 3 mm across; flight and core reach the exit	Flight and core carried down to the exit[a]
Gear	48 teeth, module 1, 50 mm over the tips, 10 mm wide, 78–88 mm from the outlet	44 teeth, module 1, 46 mm over the tips, 10 mm wide, 78–88 mm from the outlet	Four fewer teeth; same module, width, and position
Drive pinion and reduction	16 teeth; 3:1	20 teeth; 2.2:1	Lower reduction; same 32 mm centre distance
Filling	Four 7 × 4 mm slots in the closed top end (6 mm thick)	Open top end under a screw-on cap	Cap replaces the slots[a]
Reservoir	No plain section; the flight fills the bore (66 mL open volume)	Plain bore above the flight (58 mL; 79 mL open volume in all)	Plain reservoir added[a]
Cap	None; the closed top end has an M3 hole for a spindle	Screw-on, 29 mm across and 27 mm long; thread of 3.5 mm pitch, 26 mm over the crests, on the top 27 mm of the tube	Cap added[a]
Modelled with	OpenSCAD code written by an AI coding agent (May 2026)	Fusion 360, by the team (July 2026)	Remodelled by hand in conventional CAD

[a] Already present in the agent's later threaded storage auger and cap (`threaded-storage-auger-core.scad`; pull request #49, June 2026), which it modelled at the team's request. The team's Fusion 360 file of the tested auger is titled "Auger Threaded Storage".

- features that must not change (the auger's internal screw, issues #48 and #67);
- checks to run first (no interference and no floating bodies, issue #65).

Material was rarely stated, and all parts were printed in PLA.

We wrote no formal acceptance criteria but applied four in practice. The agent's checks had to pass. A team member reviewed the renders and approved a print ("We'll print and test this version", pull request #68). The printed part had to fit its neighbours and work (issue #46, pull request #49). Finally, we stopped asking for changes and froze the geometry. The type-4 auger, for example, became the main design after a bench test (issue #48). Because each request risked changing features that were already right, parts often stopped at "good enough" (discussion #39).

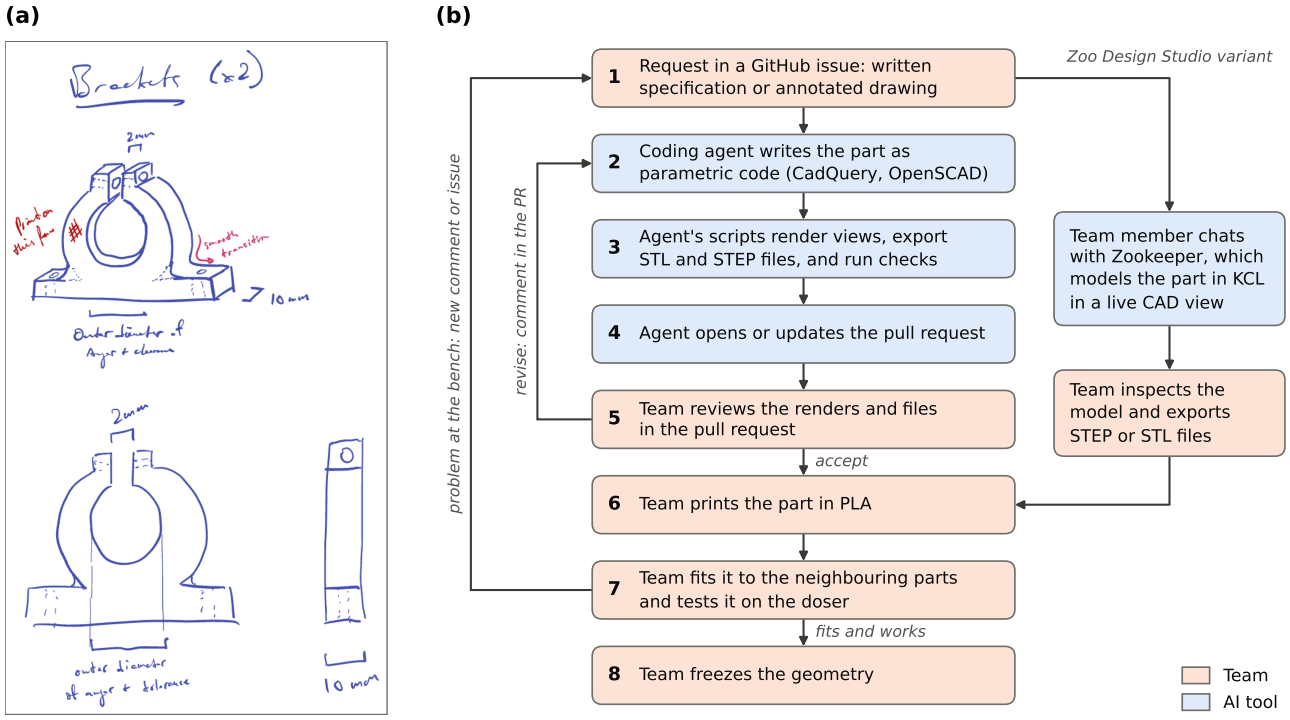


Figure S4: AI-assisted CAD workflow. (a) Sketch of the auger bracket that opened issue #46 (14 May 2026), the first part-by-part request. It gives the 2 mm clamp slot, the 10 mm thickness, the bore (auger diameter plus clearance), the face to print on, and a smooth collar-to-plate transition. The GitHub Copilot coding agent modelled it in CadQuery (pull request #47), and the same sketch was posted for the Zoo and CADSmith trials (issues #52 and #54). (b) The loop used for each part, April to June 2026. Orange boxes are team steps and blue boxes AI steps. Each Zoo Design Studio round took about an hour.

Table S4 summarizes the CAD entries of the design log. Of the 76 entries that made or changed a part model, the coding agent wrote 72, and 16 were printed. The log records 41 defects. People reviewing renders, slicer previews, or CAD caught 30 of them, printing or bench tests caught 6, and the agent's own checks caught 4. Parts that did not fit their neighbours were the most common defect (19), followed by designs built around missing or out-of-date inputs (7) and changes nobody had asked for (6). All 41 were in AI-modelled entries. The log was compiled afterwards from the repository, so these counts are lower bounds.

Table S4: CAD entries in the design log, by part family. Coding rules: an entry counts if it made or changed a 3D part model; each is coded from its log text and linked thread, and each recorded defect is credited to what first caught it. Excluded are 25 entries that added no part model or repeat another entry, and all electronics, firmware, test-rig, and design-tooling entries. Agent, an AI coding agent; team, team members in Fusion 360 or Onshape. W, worked; I, worked with issues; F, failed; S, superseded; N, not built, or printed but untested (the multi-doser); S and N entries otherwise have no print on record. Failure types follow Table 2 of the main text; family rows list the main ones. Review, a person inspecting renders, slicer previews, or CAD; test, a print or bench test; check, a mesh check or AI design review run by the agent. Revisions made for new requirements (18 entries) are not counted as failures. Some coding-agent revisions from June 2026 are missing from the log.

				Outcome					Failures		
Part family	Entries	Modelled by	Printed	W	I	F	S	N	*n*	Types	Caught by
Excavator scoop	5	agent 5	0	0	0	0	3	2	2	do not fit 2	check 1, not recorded 1
Sieve cups (A–H)	10	agent 10	0	0	0	0	9	1	0	none recorded	–
Auger	22	agent 20, team 2	10	4	3	3	10	2	11	do not fit 3, unrequested 2, mislabelled 2, malformed 2, other 2	review 6, test 4, check 1
Auger bracket	8	agent 8	3	2	0	1	3	2	4	unrequested 2, do not fit 1, bad inputs 1	review 4
Sealing cap	6	agent 6	0	0	0	0	3	3	3	do not fit 3	review 3
Tap collar	4	agent 4	1	1	0	0	3	0	2	do not fit 2	review 2
Doser module	6	agent 6	0	0	0	0	4	2	5	do not fit 3, unrequested 1, orientation 1	review 3, check 2
Mounting plate and hinge	14	agent 13, CADSmith 1	1	0	1	0	10	3	14	bad inputs 6, do not fit 5, other 3	review 12, test 2
Multi-doser	1	team 1	1	0	0	0	0	1	0	none recorded	–
Total	76	agent 72, CADSmith 1, team 3	16	7	4	4	45	16	41	do not fit 19, bad inputs 7, unrequested 6, malformed 3, mislabelled 2, failed in test 2, orientation 1, not printable 1	review 30, test 6, check 4, not recorded 1

S5 Bench noise

Protocol A takes eight balance readings with nothing moving at the start of every run. Their spread sets the smallest change that run can resolve (Fig. S5). It was 0.0 mg on every run up to 12 August 2026 and rose to tens of milligrams from 20 August. Runs are compared against their own noise floor throughout the main text.

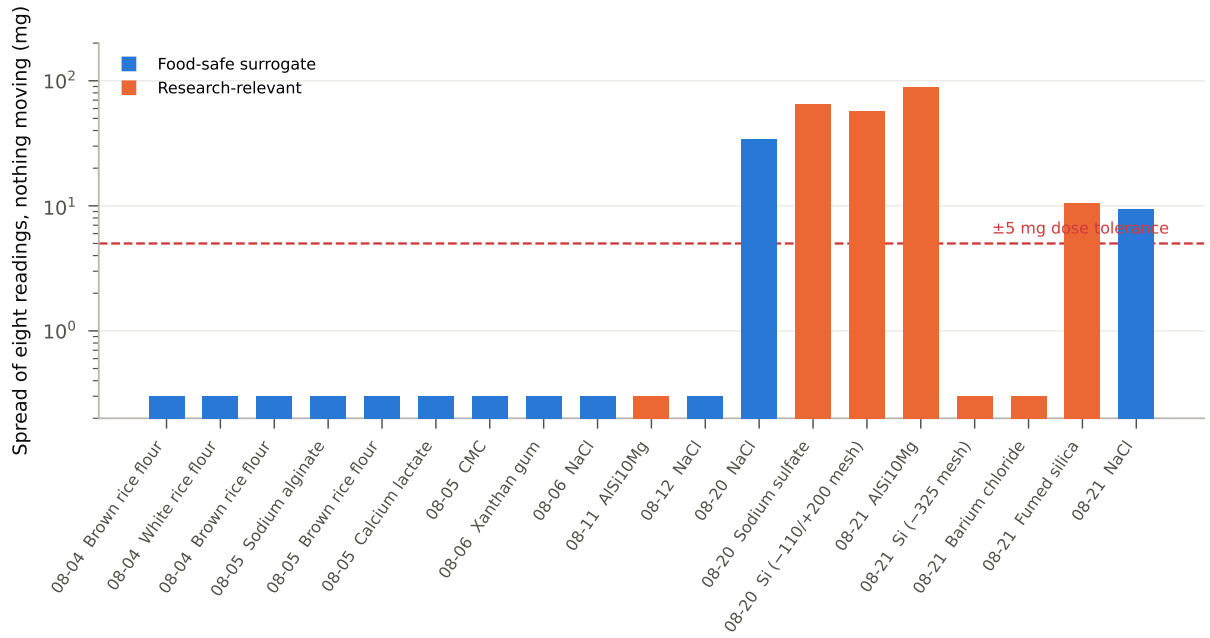


Figure S5: Balance noise floor for every first-round run: the spread (maximum minus minimum) of eight readings taken with nothing moving (protocol A), in date order. Runs whose eight readings were identical are drawn at the axis floor. The dashed line is the controller's ±5 mg stopping band.

Noise after actuation

The controller acts on readings taken 1.5 s after it moves, so we also measured the noise of those readings (Fig. S6a). For each run, we took the robust standard deviation (1.4826 times the median absolute deviation) of the change between consecutive tap-phase readings. Tapping adds only a few milligrams per cycle and the cup cannot lose mass, so on the noisier runs this scatter is balance noise. On quiet runs it is an upper bound, because it also contains the powder that each tap delivered. The noise after actuation was at most 0.9 mg in all six first-round runs with a tap phase, but 5–19 mg in five of the ten second-round runs with small doses. On those same runs, the first reading of each dose, taken at rest, scattered by at most 0.4 mg, so the stability check made before each set of doses could not detect this noise. On the noisy runs, lowering the tube to horizontal alone shifted the reading by a median of 5–16 mg, which suggests that the doser's own motion disturbed the balance.

Table S5 compares each small dose's margin to its acceptance limit with the noise of its own run. Of the 14 doses of 50 mg that finished within the limit, only one lies more than twice its run's noise inside it, and 12 lie within one noise width of the limit. Of the 16 that missed, 13 lie more than twice the noise outside it. At 200 mg, 6 of the 18 passes and 10 of the 12 failures clear twice the noise in the same way. The failures are therefore real, while most 50 mg passes cannot be told apart from noise. The controller also stops at the first reading inside its band, so on noisy runs the recorded masses lean high. All small doses come from the second round, in one fume hood with bracketed readings, so this comparison cannot be split by round or location.

Table S5: Pass/fail calls of the valid 50 and 200 mg doses compared with the balance noise after actuation in their own runs (Fig. S6). The margin is the acceptance limit minus the absolute dose error, and the noise is the robust standard deviation of the change between consecutive tap-phase readings. A call is robust when the margin exceeds twice the noise, marginal when it lies between one and two times the noise, and not resolvable otherwise. All these doses are from the second round, in the new fume hood, with bracketed readings. Per-dose values are in `paper/figures/data/post_actuation_noise_doses.csv`.

Target	Outcome	n	Robust	Marginal	Not resolvable
50 mg	within $\pm$5 mg	14	1	1	12
50 mg	outside $\pm$5 mg	16	13	0	3
200 mg	within $\pm$10 mg	18	6	3	9
200 mg	outside $\pm$10 mg	12	10	0	2

S6 Human tasks in the test campaign

Table S6 lists every task a person did in the test campaign, when it is needed, how often it happened, and what could automate it. Nobody acted between doses. Every run was started unattended, so the firmware's 137 requests to empty the cup continued at once, and each dose was weighed from its own tare with earlier powder left in the cup. The human work fell between runs (filling, mounting, and positioning an auger, and emptying the cup) and on faults, most of which involved the balance or its surroundings. Each powder had its own printed auger, labelled by hand and stored filled, with the cap on and tape over the outlet. Counts taken from the issue threads are lower bounds.

S7 Closed-loop dose record

Table S7 accounts for every closed-loop dose attempt in both test rounds, including the attempts we excluded and the combinations of powder and target that were never run. A dose is valid when its run passed quality screening, the dose ended in a defined controller state, and the auger or solenoid moved. Two 50 mg doses, on calcium lactate and on coarse silicon, were first excluded because the firmware logged no auger revolution, but their serial logs show a short continuous spin that delivered 62 and 55 mg, so we count them as valid overshoots. Table S8 summarizes the valid doses by powder and target mass.

Fig. S7 shows balance readings during representative doses of a fast and a slow powder. Doses that finished within the limit spent about half their time in the tap phase (median 46% at 50 mg, 50% at 200 mg, and 53% at 1 g). On the free-flowing powders, one 45° auger turn delivered a median of 30 mg, against 2 mg on the slow powders. The last step of a 50 mg dose on a free-flowing powder added a median of 46 mg (range 8–100 mg), more than the whole $\pm$5 mg acceptance band, so a dose could step right over it. On the slow powders the last step added at most 9 mg, and doses stopped near the lower edge of the band. In 19 of the 60 valid 50 and 200 mg doses, the controller began with a brief continuous spin that delivered up to 103 mg. Its bulk phase started whenever the tared reading sat a fraction of a milligram below zero, because the threshold for that phase equalled the target. Per-dose phase times and final increments are in `paper/figures/data/dose_phase_summary.csv`.

Table S6: Human tasks in the test campaign (37 recorded runs and 137 closed-loop doses in two rounds, 4–21 August and 3–15 September 2026). Counts come from the raw serial logs where the firmware recorded the task, and otherwise from the issue threads and run notes. Every run was started unattended, so the firmware's requests to the operator continued at once and no one acted between doses. Counts and sources for each task are in `paper/figures/data/operator_tasks.csv`.

Task	When needed	How often in the campaign	What could automate it
Empty the collection cup	Every dose	Asked for before all 137 closed-loop doses and before 57 protocol C tilts. No request was answered, because every run was unattended: earlier powder stayed in the cup, and each dose was weighed from its own tare.	A second cup position or a cup changer
Tare the balance		Done by the firmware before each dose. From 3 Sep it also checked each tare and subtracted any residue (21 of 89 had not taken). Before that, 2 refused tares read leftover powder as doses of 7.54 and 1.54 g.	An auto-tare check (in the firmware from 3 Sep)
Fill and label an auger; top it up	Every run or powder change	At least 14 of the 16 printed augers were filled: one per powder, plus a second for brown rice flour. Top-ups were recorded at least 3 times. The firmware asked 15 times whether the tube was empty, and each was answered "keep" automatically.	A cap-and-refill station with fill-level sensing
Load an auger: mount it, seat its gear, remove the outlet tape		The operators reported 37 loads; the order of the runs alone needed 29. Tape was left on the outlet once (5 Aug); the feed check and a camera frame caught it.	A cartridge changer with a tape-free outlet seal
Position the outlet over the cup; level the balance		Done by eye, with no fixture. Levelling was recorded 3 times, after moves of the rig. One run was excluded because the outlet missed the cup (8 Sep).	A kinematic mount that fixes the outlet over the cup
Check the outlet and feed before a run		A scripted feed check (five auger revolutions) is recorded for at least 30 of the 37 runs, 2 of which went no further, and camera frames were checked remotely. It also filled the outlet flights when hand-turning after loading was skipped (4 times recorded).	The scripted check and camera (both used); an outlet sensor
Empty and clean the cup; dispose of powder; wipe up spills		By hand between runs, but powder stayed in the cup across a run or a powder change at least 3 times. A beaker broke, and 3 more vessels were tried before one fitted. Drying after washing delayed 2 runs by 50 and 70 min.	Disposable cups, a funnel, or a pan-clearing step
Clear a feed fault	On a fault	Brown rice flour was moved into a second auger (5 Aug), and a blocked test auger fitted for other work was removed (15 Sep). Barium chloride caked after 3 weeks in its auger and was not cleared (10 Sep; run excluded).	Tapping before dosing; sealed, dry storage
Repair the mechanism		The tilt servos seized (11 Aug; run excluded; repaired next day), a hinge screw backed out (11 Aug), and the tap collar caught under the mounting plate (19 Aug).	A tilt sensor; locknuts
Restore the balance		Found off or in standby 3 times. Error 1 resisted zeroing by hand and was cleared remotely (19–20 Aug). After the last run, drift was traced to powder under the dust plate.	Automatic power-on and a standby check at start-up
Restore the control link; confirm that a run ended		The Pi was power-cycled by hand once (7 Aug), and 2 network drops cleared on their own. The operator asked 9 times whether a run had ended, because the AI agent's report had stalled.	A network watchdog; status messages from the rig
Move the rig or wait for a quiet room		Moved 4 times between the open bench and two fume hoods. Dose runs were postponed 5 times, 4 of them for Si $-110/+200$, which finally ran in an empty lab in the evening.	An enclosure on a vibration-isolated base; the stability check (used)

Table S7: Accounting for every closed-loop dose attempt in both test rounds, by powder and target mass. *Attempts* counts every dose the controller started. A dose is *valid* when its run passed screening, it ended in a defined controller state, and the auger or solenoid moved (a continuous spin in the bulk phase counts even when it was too short for the firmware's 1 s clock to log a revolution). *Within limit* counts valid doses inside the acceptance limit (±10% at 50 mg, ±5% at 200 mg and 1 g). The firmware columns give the controller's own end label for each valid dose. *ok*: inside its ±5 mg stopping band; *over*: more than 5 mg above the target; *stall*: the reading stopped rising; *budget*: 200 cycles of one phase; *time*: 900 s. The controller stopped on its own ±5 mg band, whereas doses are scored against the acceptance limit, so the two counts differ. At 1 g, for example, 9 of the 18 doses labelled stalled were within the ±50 mg limit. The two indented rows split the 1 g total by powder. The 24 doses within the limit are every dose on the seven fastest-flowing powders dosed (24 of 24): AlSi10Mg, coarse silicon, sodium sulfate, calcium lactate, xanthan gum, NaCl, and CMC. The 15 valid doses on the other four (white rice flour, sodium alginate, fine silicon, and brown rice flour) all missed it. The last column gives the reason for each excluded attempt, with the number of attempts, or why a combination was not run. One valid 200 mg calcium lactate dose was reconstructed from its balance trace and camera record after the serial link failed. Powders are in the order of Table 1 of the main text. Per-dose records are in `paper/figures/data/doses_all.csv`.

			Within	Firmware end label (valid doses)					Excluded attempts,
Powder	Attempts	Valid	limit	ok	over	stall	budget	time	or why not run
50 mg target (acceptance limit ±5 mg)									
AlSi10Mg	3	3	1	1	2	0	0	0	
Si (−110/+200 mesh)	3	3	2	2	1	0	0	0	
Sodium sulfate	3	3	0	0	3	0	0	0	
Calcium lactate	3	3	1	1	2	0	0	0	
Barium chloride	3	0	–	–	–	–	–	–	powder caked in the tube (3)
Xanthan gum	3	3	2	2	1	0	0	0	
NaCl	9	3	2	2	1	0	0	0	balance could not be read (6)
CMC	6	3	3	3	0	0	0	0	outlet misaligned (3)
White rice flour	3	3	3	3	0	0	0	0	
Sodium alginate	3	3	0	0	0	2	0	1	
Si (−325 mesh)	3	3	0	0	0	3	0	0	
Brown rice flour	0	–	–	–	–	–	–	–	not run: delivered almost nothing in round 1
Fumed silica	0	–	–	–	–	–	–	–	not run: delivered almost nothing in round 1
All powders	42	30	14	14	10	5	0	1	balance could not be read (6); outlet misaligned (3); powder caked in the tube (3)
200 mg target (acceptance limit ±10 mg)									
AlSi10Mg	3	3	3	3	0	0	0	0	
Si (−110/+200 mesh)	3	3	2	2	1	0	0	0	
Sodium sulfate	3	3	2	1	2	0	0	0	
Calcium lactate	3	3	2	1	0	1	1	0	
Barium chloride	3	0	–	–	–	–	–	–	powder caked in the tube (3)
Xanthan gum	3	3	3	3	0	0	0	0	
NaCl	9	3	2	2	1	0	0	0	balance could not be read (6)
CMC	6	3	1	1	0	0	0	2	outlet misaligned (3)
White rice flour	3	3	3	3	0	0	0	0	
Sodium alginate	3	3	0	0	0	1	0	2	
Si (−325 mesh)	3	3	0	0	0	3	0	0	
Brown rice flour	0	–	–	–	–	–	–	–	not run: delivered almost nothing in round 1

continued on next page

Powder	Attempts	Valid	Within limit	Firmware end label (valid doses) ok	over	stall	budget	time	Excluded attempts, or why not run
Fumed silica	0	–	–	–	–	–	–	–	not run: delivered almost nothing in round 1
All powders	42	30	18	16	4	5	1	4	balance could not be read (6); outlet misaligned (3); powder caked in the tube (3)
1 g target (acceptance limit ±50 mg)									
AlSi10Mg	6	3	3	2	1	0	0	0	tilt servo failed (3)
Si (−110/+200 mesh)	3	3	3	3	0	0	0	0	
Sodium sulfate	3	3	3	3	0	0	0	0	
Calcium lactate	3	3	3	0	0	3	0	0	
Barium chloride	3	0	–	–	–	–	–	–	powder caked in the tube (3)
Xanthan gum	3	3	3	0	0	3	0	0	
NaCl	7	6	6	5	1	0	0	0	demonstration run (1)
CMC	3	3	3	0	0	3	0	0	
White rice flour	3	3	0	0	0	0	3	0	
Sodium alginate	3	3	0	0	0	0	3	0	
Si (−325 mesh)	6	6	0	0	0	6	0	0	
Brown rice flour	9	3	0	0	0	3	0	0	nothing delivered, auger suspected (6)
Fumed silica	0	–	–	–	–	–	–	–	not run: delivered almost nothing in round 1
All powders	52	39	24	13	2	18	6	0	nothing delivered, auger suspected (6); tilt servo failed (3); powder caked in the tube (3); demonstration run (1)
seven fastest-flowing	28	24	24	13	2	9	0	0	tilt servo failed (3); demonstration run (1)
other four dosed	21	15	0	0	0	9	6	0	nothing delivered, auger suspected (6)
NaCl at 0.5 and 2 g	3	0	–	–	–	–	–	–	demonstration run (3)
All attempts	139	99	56	43	16	28	7	5	balance could not be read (12); powder caked in the tube (9); nothing delivered, auger suspected (6); outlet misaligned (6); demonstration run (4); tilt servo failed (3)

Table S8: Closed-loop dose summary for every valid dose (protocol G at 1 g; protocol H at 50 and 200 mg). Error is delivered minus requested mass. *Within limit*: doses inside the acceptance limit (±10% at 50 mg, ±5% at 200 mg and 1 g). *Firmware ok*: doses the controller ended inside its own ±5 mg stopping band. Round 1: August 2026, balance read on its stability flag; round 2: September 2026, bracketed readings. Per-dose records, including invalid doses and the reason each was excluded, are in `paper/figures/data/doses_all.csv`.

Powder	Target	Round	*n*	Mean error (mg)	SD (mg)	Within limit	Firmware ok	Median time (s)
AlSi10Mg	50 mg	2	3	+21.5	26.7	1/3	1/3	15
Si (−110/+200 mesh)	50 mg	2	3	+2.5	4.1	2/3	2/3	10
Sodium sulfate	50 mg	2	3	+28.9	15.7	0/3	0/3	10
Calcium lactate	50 mg	2	3	+10.6	11.2	1/3	1/3	12
Xanthan gum	50 mg	2	3	+17.8	36.4	2/3	2/3	37
NaCl	50 mg	2	3	+4.2	4.8	2/3	2/3	17
CMC	50 mg	2	3	-2.2	2.4	3/3	3/3	211
White rice flour	50 mg	2	3	-4.7	0.6	3/3	3/3	136
Sodium alginate	50 mg	2	3	-19.6	8.3	0/3	0/3	561
Si (−325 mesh)	50 mg	2	3	-50.0	0.1	0/3	0/3	31
AlSi10Mg	200 mg	2	3	-0.3	2.7	3/3	3/3	214
Si (−110/+200 mesh)	200 mg	2	3	+9.8	19.8	2/3	2/3	34

Powder	Target	Round	n	Mean error (mg)	SD (mg)	Within limit	Firmware ok	Median time (s)
Sodium sulfate	200 mg	2	3	+10.8	17.7	2/3	1/3	30
Calcium lactate	200 mg	2	3	-16.5	16.6	2/3	1/3	464
Xanthan gum	200 mg	2	3	+2.9	0.6	3/3	3/3	89
NaCl	200 mg	2	3	+2.8	9.2	2/3	2/3	53
CMC	200 mg	2	3	-22.6	23.4	1/3	1/3	903
White rice flour	200 mg	2	3	-4.9	0.1	3/3	3/3	766
Sodium alginate	200 mg	2	3	-101.5	73.3	0/3	0/3	903
Si (−325 mesh)	200 mg	2	3	-199.7	0.5	0/3	0/3	31
AlSi10Mg	1 g	2	3	+1.9	6.6	3/3	2/3	65
Si (−110/+200 mesh)	1 g	2	3	+0.2	2.8	3/3	3/3	128
Sodium sulfate	1 g	2	3	-4.2	1.1	3/3	3/3	86
Calcium lactate	1 g	1	3	-26.5	9.9	3/3	0/3	263
Xanthan gum	1 g	1	3	-32.9	9.6	3/3	0/3	193
NaCl	1 g	1	6	-1.9	5.3	6/6	5/6	186
CMC	1 g	1	3	-43.2	5.4	3/3	0/3	668
White rice flour	1 g	1	3	-137.9	23.5	0/3	0/3	822
Sodium alginate	1 g	1	3	-291.5	18.2	0/3	0/3	840
Brown rice flour	1 g	1	3	-999.1	0.9	0/3	0/3	7
Si (−325 mesh)	1 g	1,2	6	-999.5	0.7	0/6	0/6	8

S8 Composition error from dosing error

The acceptance limits apply to each feedstock in a blend, and a feedstock built from several passing doses of 100 mg or more is also within ±5%. Let feedstock f have mass m_f, relative mass error ε_f, and element mass fractions c_{fk}, and let M_k be the atomic weight of element k. With $n_k = \sum_f m_f c_{fk}/M_k$ and $N = \sum_k n_k$, the mole fraction $x_k = n_k/N$ changes to first order by

$$\delta x_k = \sum_f S_{kf}\,\varepsilon_f, \qquad S_{kf} = \frac{m_f}{N}\left(\frac{c_{fk}}{M_k} - x_k \sum_j \frac{c_{fj}}{M_j}\right). \tag{1}$$

Mass fractions follow by setting every M to one. For pure elements, $\delta x_i = x_i(\varepsilon_i - \sum_j x_j \varepsilon_j)$. If every error sits at a common limit L with the least favourable signs, $|\delta x_i| = 2x_i(1-x_i)L$. For an equimolar five-component blend ($x_i = 0.2$, $L = 5\%$), this is 1.6 atomic percent, or 1.65 without the linear approximation. If the errors are independent and uniform within the limits, the standard deviation is 0.52 atomic percent.

Table S9 applies Eq. 1 to batches of 10, 100, and 500 g, the range planned for our alloy work. Batch size only decides which limit applies, so the dosing columns barely change with it. With uniform errors, each element of the equimolar blend stays within one atomic percent in 97% of simulated blends. The measured errors of accepted doses give a standard deviation of 0.40 atomic percent, and 0.08 atomic percent for the three research-relevant powders. The aluminium-rich blends we plan to make, such as Al-5Mn-2Cr-2Zr-0.5Ti, stay near 0.4 atomic percent even in the worst case. The Al-24Si test blend behaves like the equimolar blend.

The balance also records each delivered mass, so the composition actually dosed is known to within the balance uncertainty. For a 10 g batch, this is 0.002 atomic percent at 0.1 mg and 0.08 atomic percent at 5 mg. A dose that misses its target can then be corrected, by re-targeting later feedstocks to the recorded masses or topping up earlier ones. Feedstock chemistry adds an error of similar size: supplier ranges for the silicon and AlSi10Mg powders put the 100 g Al-24Si blend anywhere from 23.6 to 24.8 wt% Si. Processing can add far more, and directed energy deposition from elemental powders has shifted compositions by 5–10 atomic percent.[1,2] The dosing limits are therefore suited to screening compositions. They do not by themselves fix the composition of an atomized or printed alloy, which still has to be measured.

Table S9: Composition error added by dosing, given as the largest absolute error of any element in the blend. Each feedstock is dosed within its acceptance limit (±10% of its mass below 100 mg, ±5% at or above). *Worst*: every feedstock at its limit, with the signs that give the largest error. *SD*: independent errors, either uniform within the limit (SD equal to the limit divided by $\sqrt{3}$) or with the measured root-mean-square error of accepted doses (7.0%, 2.1%, and 2.2% at 50 mg, 200 mg, and 1 g). *Recorded masses*: SD when the composition is calculated from the recorded masses, each with a standard uncertainty of 0.1 mg or 5 mg. Al-5Mn-2Cr-2Zr-0.5Ti (wt%, Zr added as an Al-10Zr master alloy) and Al-24Si-0.1Mg (made from Al, Si, and AlSi10Mg for atomization trials) are blends from our alloy-discovery workflow. Values for every element are in `paper/figures/data/composition_error.csv`.

			Dosing within the limits				Recorded masses	
Blend	Batch (g)	Smallest feedstock (g)	Worst (at%)	Worst (wt%)	SD, uniform (at%)	SD, measured (at%)	0.1 mg (at%)	5 mg (at%)
Equimolar AlCoCrFeNi	10	1.07 (Al)	1.65	1.84	0.52	0.40	0.002	0.08
	100	10.7 (Al)	1.65	1.84	0.52	0.40	<0.001	0.008
	500	53.4 (Al)	1.65	1.84	0.52	0.40	<0.001	0.002
Al-5Mn-2Cr-2Zr-0.5Ti	10	0.050 (Ti)	0.41	0.73	0.13	0.10	<0.001	0.05
	100	0.500 (Ti)	0.40	0.71	0.13	0.10	<0.001	0.005
	500	2.50 (Ti)	0.40	0.71	0.13	0.10	<0.001	<0.001
Al-24Si-0.1Mg	10	2.03 (Si)	1.55	1.58	0.54	0.41	<0.001	0.04
	100	20.3 (Si)	1.55	1.58	0.54	0.41	<0.001	0.004
	500	102 (Si)	1.55	1.58	0.54	0.41	<0.001	<0.001

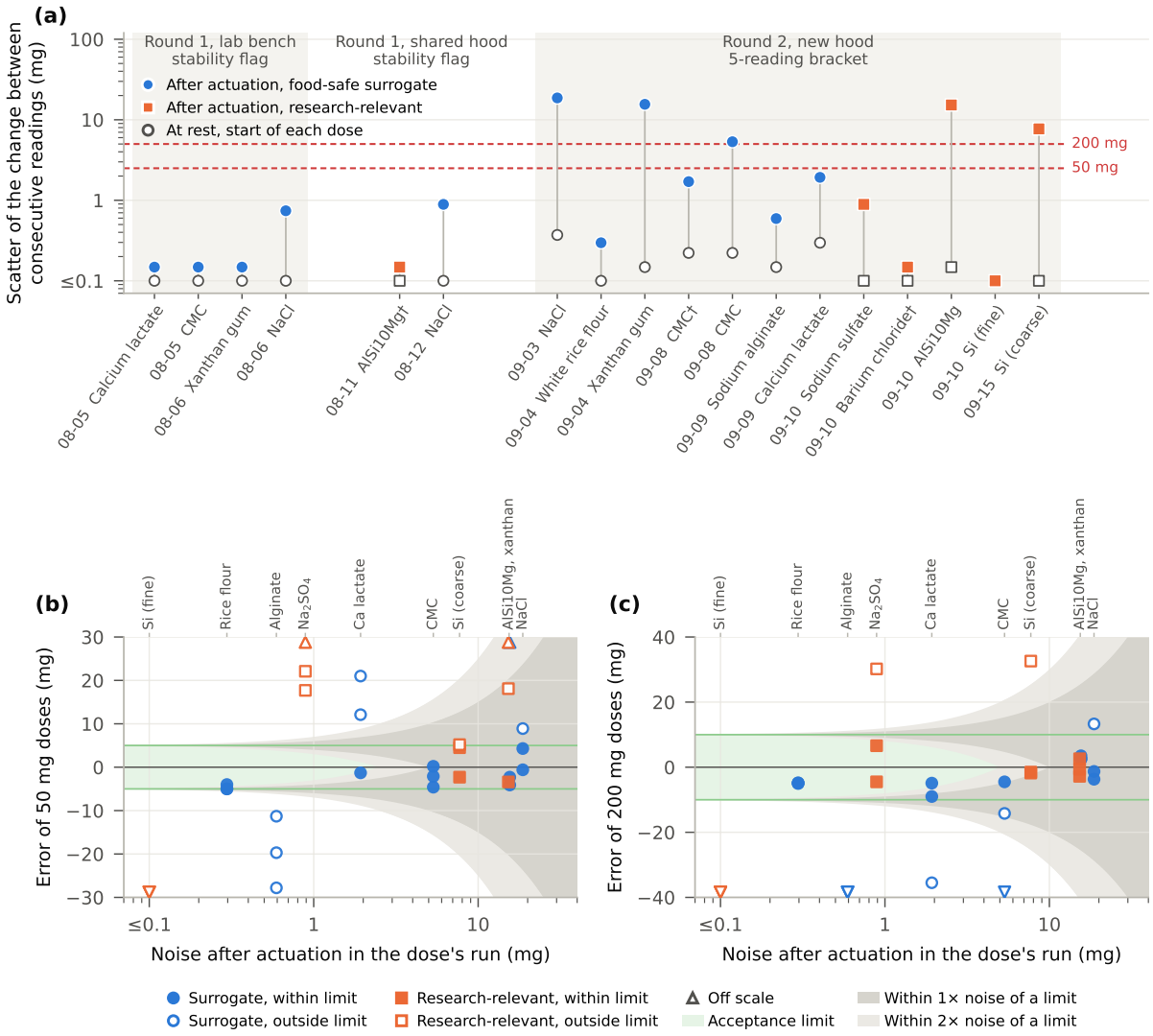


Figure S6: Balance noise after actuation and the pass/fail calls of small doses. (a) Filled markers: noise after actuation for every run with a closed-loop tap phase, the robust standard deviation of the change between consecutive tap-phase readings, each taken 1.5 s after two solenoid taps with the tube horizontal. Open markers apply the same statistic at rest, to the first reading of each dose minus its tare reading. Fine silicon and barium chloride never reached the tap phase, so their values come from fine-phase steps that delivered nothing. Values below 0.1 mg are drawn at 0.1 mg, and † marks runs excluded from the dose analysis. Above the dashed lines, no 50 mg or 200 mg dose can pass by more than twice the noise. (b, c) Error of each valid 50 and 200 mg dose against the noise of its run (named along the top). Light and dark grey zones lie within two and one times the noise of a limit. Markers as in Fig. 5 of the main text.

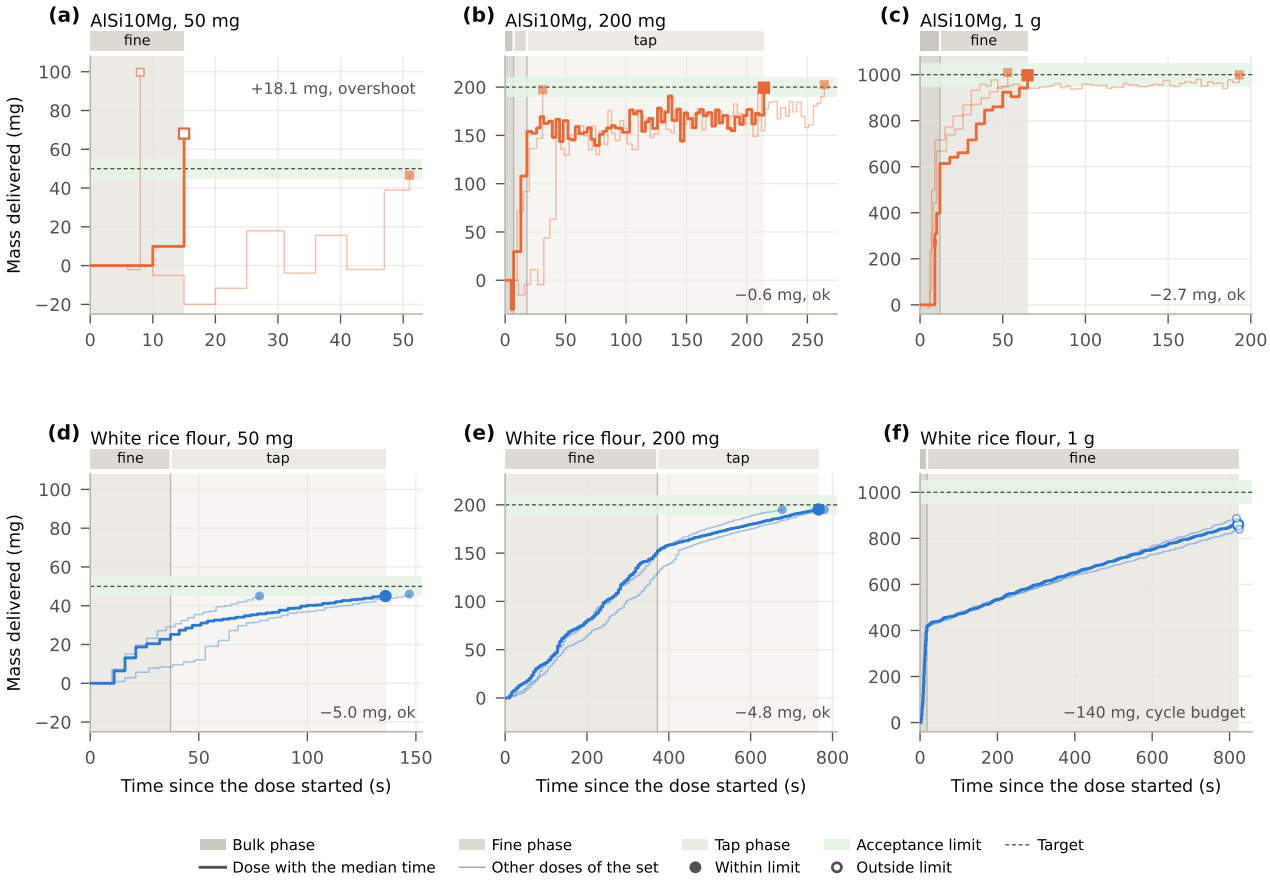


Figure S7: Balance readings during closed-loop doses of a fast, free-flowing powder (a–c, AlSi10Mg) and a slow one (d–f, white rice flour) at 50 mg, 200 mg, and 1 g. The thick line is the dose with the median time of its set of three, and thin lines are the other two. Shading and the bar above each panel mark the phases of the thick-line dose: bulk (continuous rotation, tube at 45°), fine (45° auger turns, tube at 22.5°), and tap (pairs of taps, tube horizontal). The green band is the acceptance limit and the dashed line the target. Markers are filled within the limit and open outside it. Time axes differ. In (a), the second 45° turn added 58 mg, more than the whole target. In (b), a first reading just below zero triggered a brief bulk spin. Tap-phase readings then ranged over 140–191 mg, and the dose stopped on the first reading inside the ±5 mg band. Panels (a–e) are from September 2026, and (f) is from 4 August 2026.

S9 Test protocols as recorded

Table 3 of the main text specifies the eight test protocols in tube angles and auger units. Table S10 gives the same protocols as the raw records store them, and counts the runs and records behind each one, so that a reader can match any row of the raw CSV files to its protocol.

The test battery's raw records store tilt as a servo setting (0, 45, or 90) rather than as an angle. The tilt plate is geared 2:1, so these settings are tube angles of 0°, 22.5°, and 45° above horizontal, and 45° is the steepest tilt reached in any test. The main text and its figures report tube angles; the conversion is made once, in `paper/figures/make_data_figures.py`.

Table S10: Test protocols as they appear in the raw records, and how many records each produced. Table 3 of the main text gives the same protocols in tube angles. Here tilt is the servo setting the records store (0, 45, or 90), which the 2:1 tilt gear turns into tube angles of 0°, 22.5°, and 45°. The firmware and the CSV files identify the protocols by the same letters, in a field named `block`. A record is one trial for protocols A–E (for example, one auger revolution in C, or one re-feed or one tap in E) and one dose attempt for G and H. *From screened runs* counts the records from runs that passed screening; for G and H it counts the doses scored in the main text, after the exclusions listed in Table S7. Protocol D's timing loop turned the auger 15–54% further than the three revolutions commanded (Experimental). Every run requesting protocol F logged a skip because the haptic driver did not respond. Two demonstration doses on 1 September 2026 that have no run record are left out. Counts are computed from `runs_all.csv`, `trials.csv`, and `doses_all.csv` in `paper/figures/`.

Protocol		Settings in the raw records	Runs asking		Runs with	Records	From screened
			Round 1	Round 2	records		runs
A	Balance baseline	tilt 45; 8 reads; no actuator	19	0	19	152	112
B	Static hold	tilt 0, 45, 90; 15 s hold each	19	0	19	57	42
C	Rotation	tilt 0, 45, 90; 6 × 360° at 30 rpm	19	0	19	342	252
D	Speed sweep	tilt 45; 15, 45, 90 rpm; 3 rev commanded; poll 250 ms	19	0	19	57	42
E	Tap	tilt 0, 45; 8 × (360° re-feed + one 60 ms tap)	19	0	19	592	448
F	Vibration	tilt 0, 45; as E with 3 ERM bursts	11	0	0	0	0
G	Closed-loop, 1 g	3 doses; target 1000 mg	12	7	19	53	39
H	Closed-loop, small	3 doses each; targets 50 and 200 mg	0	14	14	84	60

S10 Run inventory

Table S11 lists every run recorded in the two test rounds, including runs that were aborted, postponed, or excluded, so that each number in the main text can be traced to the run it came from.

Table S11: Every recorded run in both test rounds, in date order. *Protocols* lists the test protocols the run requested (Table 3 of the main text; protocol F never produced data); *Doses* is the number of closed-loop doses attempted. Locations: an open laboratory bench (to 6 August 2026), a shared fume hood (11 August to 3 September), and a second fume hood (from 3 September). Runs that failed screening are listed here and in `paper/figures/data/runs_all.csv` with the reason; the Experimental section describes how runs were chosen for each figure.

Start (UTC)	Round	Powder	Protocols	Location	Doses	Screening
2026-08-04 20:43	1	Brown rice flour	ABCDEFG	open bench	3	excluded: nothing delivered, auger suspected
2026-08-04 21:14	1	White rice flour	ABCDEFG	open bench	3	passed
2026-08-04 22:49	1	Brown rice flour	ABCDEFG	open bench	3	excluded: nothing delivered, auger suspected
2026-08-05 14:57	1	Sodium alginate	ABCDEFG	open bench	3	passed
2026-08-05 18:53	1	Brown rice flour	ABCDEFG	open bench	3	passed (slow delivery)
2026-08-05 20:00	1	Calcium lactate	ABCDEFG	open bench	3	passed

Start (UTC)	Round	Powder	Protocols	Location	Doses	Screening
2026-08-05 21:12	1	–	–	open bench	0	aborted before testing
2026-08-05 21:52	1	CMC	ABCDEFG	open bench	3	passed
2026-08-06 14:02	1	Xanthan gum	ABCDEFG	open bench	3	passed
2026-08-06 14:51	1	NaCl	ABCDEFG	open bench	3	passed
2026-08-11 18:30	1	AlSi10Mg	ABCDEFG	fume hood 1	3	excluded: tilt servo failed
2026-08-12 21:51	1	NaCl	ABCDEFG	fume hood 1	3	passed
2026-08-20 17:56	1	NaCl	ABCDE	fume hood 1	0	excluded: deliberate bench stress test
2026-08-20 19:22	1	Sodium sulfate	ABCDE	fume hood 1	0	passed
2026-08-20 21:48	1	Si (−110/+200 mesh)	ABCDE	fume hood 1	0	passed
2026-08-21 15:13	1	AlSi10Mg	ABCDE	fume hood 1	0	passed
2026-08-21 16:18	1	Si (−325 mesh)	ABCDEG	fume hood 1	3	passed (slow delivery)
2026-08-21 17:07	1	Barium chloride	ABCDE	fume hood 1	0	passed
2026-08-21 18:27	1	Fumed silica	ABCDE	fume hood 1	0	excluded: nothing delivered, outlet not checked
2026-08-21 19:20	1	NaCl	ABCDE	fume hood 1	0	passed
2026-09-03 16:35	2	NaCl	H	fume hood 1	6	excluded: balance could not be read
2026-09-03 17:04	2	NaCl	H	fume hood 1	6	excluded: balance could not be read
2026-09-03 21:40	2	NaCl	H	fume hood 2	6	passed
2026-09-04 00:18	2	NaCl	G	fume hood 2	1	demonstration only
2026-09-04 19:00	2	White rice flour	H	fume hood 2	6	passed
2026-09-04 20:37	2	Xanthan gum	H	fume hood 2	6	passed
2026-09-08 17:27	2	CMC	H	fume hood 2	6	excluded: outlet misaligned
2026-09-08 19:42	2	CMC	H	fume hood 2	6	passed
2026-09-09 18:53	2	Sodium alginate	H	fume hood 2	6	passed
2026-09-09 22:52	2	Calcium lactate	H	fume hood 2	5	passed
2026-09-10 13:39	2	Sodium sulfate	GH	fume hood 2	9	passed
2026-09-10 17:08	2	Barium chloride	GH	fume hood 2	9	excluded: powder caked in the tube
2026-09-10 19:44	2	AlSi10Mg	GH	fume hood 2	9	passed
2026-09-10 21:29	2	Si (−325 mesh)	GH	fume hood 2	9	passed (slow delivery)
2026-09-10 23:19	2	–	–	fume hood 2	0	postponed: failed the stability check
2026-09-15 02:55	2	Si (−110/+200 mesh)	GH	fume hood 2	9	passed
2026-09-15 16:42	2	NaCl	G	fume hood 2	1	demonstration only

S11 AI usage log

In line with *Digital Discovery*'s guidance on the use of LLMs, all AI interactions are preserved publicly in the repository:

- **CAD and code generation:** every coding-agent session is recorded as a pull request containing the prompt (issue text and review comments), the generated CAD code, rendered views, and the human review thread. The raw session logs that GitHub Actions keeps expire after 90 days: those from April to early July 2026 had expired by October 2026, and the later logs and the Claude Code session transcripts were downloaded then and are part of the archived release. Table S12 lists each AI tool, the months it was used, and the models the records name.
- **Design reviews and literature queries:** Edison Scientific query scripts, task identifiers, raw answers, and extracted references are archived with the corresponding branches; the consolidated bibliography, with the verification status of each entry, is `paper/references_validation.md`.
- **Text-to-CAD evaluations:** CADSmith run artifacts and the Zoo Design Studio (Zookeeper agent) session transcripts and notes are preserved on their branches.
- **Analysis and writing:** the data-analysis scripts behind every figure, and the AI-drafted revisions of this manuscript, are in `paper/` with their review threads.

- **Qualitative observations:** the team's running journal of AI behaviour, good and bad, is discussion #39 in the repository.

Table S12: AI tools used in this project, the months of 2026 in which the repository records their use, and the model names it records. Model names for the coding agents come from tags in the requests (`@copilot+`*model*), the history of the Claude Code workflow file, and commit trailers. Requests without a tag used a default model that the record does not show. AI tools tried for circuit-board layout (issues #82, #94, and #95) are not listed. Links and commit hashes for every entry are in `paper/figures/data/ai_tools_timeline.csv`.

Tool	Role in this project	Months (2026)	Models recorded	Evidence
GitHub Copilot coding agent	Modelled the AI-made parts as CadQuery or OpenSCAD code in pull requests, with renders, STL and STEP files, and checks; also documents and early drafts	Apr–Jun; once in Jul	Claude Opus 4.7 and Opus 4.8 (most tagged requests); Haiku 4.5, Sonnet 4.6, and Fable 5 (a few)	Model tags in 233 of 300 requests; PRs #47, #49, #51, #66
Claude, local session	First auger, written at the April hackathon (OpenSCAD)	Apr	Claude Sonnet 4.6	Commit trailers in PR #16
Claude Code (GitHub Action)	Test campaign, firmware, analysis, figures, and writing; Onshape and text-to-cad recreations of the doser	30 Jun–Oct	Claude Opus 4.8, Fable 5, Opus 5, Opus 5.5, and Fable 5.1	History of `claude.yml`; commit trailers
CADSmith	Trial runs on a chassis, the mounting plate and baseplate, and the whole doser	May–Jun	Claude Opus 4.7; Sonnet 4.5 with an Opus 4.5 judge; Opus 4.5 for all agents	PR #59; CADSmith fork PRs #2 and #6
Zoo Text-to-CAD API	KCL and STEP parts from text prompts, called by the coding agent	May–Jun	Zoo `kcl` model, version 0.1.0	PRs #7, #49, and #110
Zoo Design Studio (Zookeeper agent)	Modelling by chat in a live CAD view, for example the servo pinion and the final tap collar	May–Jul (mainly Jun)	Zookeeper; underlying model not recorded	Issues #92 and #104; PR #66; discussion #39
Edison Scientific	Literature and prior-art searches; reviews of designs, data, and drafts	Apr–Oct	Job types only: Literature, Analysis, and Precedent	Query scripts and task records
GitHub Copilot code review	Automatic pull-request reviews; six suggested fixes accepted	Apr–Jun	Not recorded	Actions log; PR #35

Fig. S8 shows how agent use was spread over the project. Team members addressed the two coding agents 754 times between April and September 2026, and the agents authored 789 of the 877 commits (90%). The counts cover all work in the repository: CAD, firmware, data analysis, and writing. The team moved from the GitHub Copilot coding agent to Claude Code at the end of June 2026.

References

[1] M. Moorehead, K. Bertsch, M. Niezgoda, C. Parkin, M. Elbakhshwan, K. Sridharan, C. Zhang, D. Thoma and A. Couet, *Materials & Design*, 2020, **187**, 108358.

[2] Y. Chen, X. Zhang, M. M. Parvez and F. Liou, *Materials*, 2020, **13**, 3562.

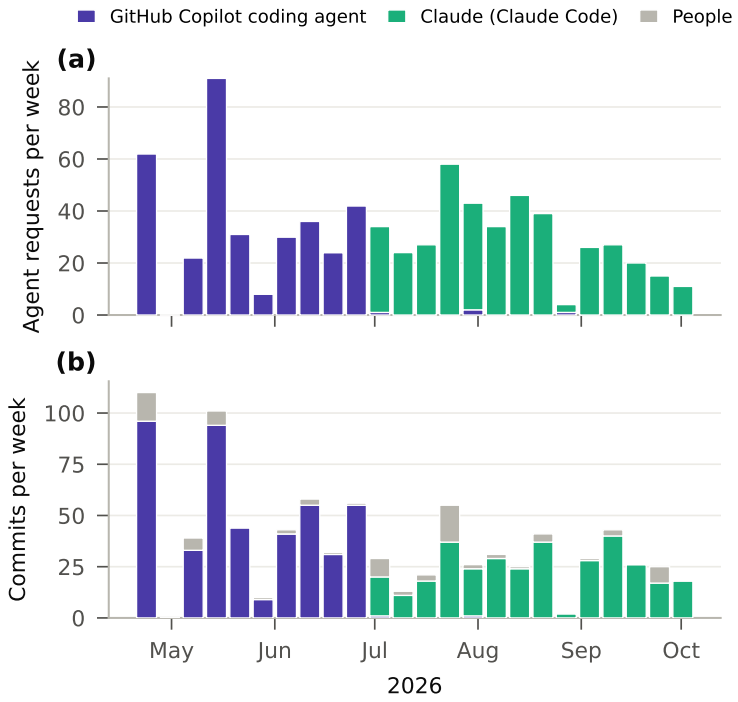


Figure S8: Generative-AI agent activity per week over the project. (a) Requests to an agent: comments and issue or pull-request descriptions written by a team member that address `@copilot` or `@claude`, plus pull requests the Copilot coding agent opened when assigned an issue. Most such requests started one agent session, so the count is an approximate measure of use. (b) Commits on all branches by author; commits that were later rebased onto another branch are counted once. Counts are from the public repository history (`paper/figures/data/build_ai_usage.py`).